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Project Title:	Electric Field Control of Flames		
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	NOTE this is a successor agreement to "Applications of Electric Fields in Microgravity Combustion Science (NNX07AB55A)," per D. Stocker, NASA Glenn Research Center. This joint project between the University of California, Irvine and Imperial College, London explores the use of large electric fields in combustion for the purpose of improving the performance of energy conversion systems. As an actuator, electric fields can affect flame stability and soot formation; while as a sensing device, the electrical response at saturation is a flame characteristic. In addition to preparing for the proposed ACME E-FIELD experiments aboard the ISS, we are currently studying quenched jet flames, partially-admixed air/fuel jet flames, and their associated dangers. This report summarizes recent findings. Specific activities include: 1) Characterizing the high voltage power supplies planned for experiments aboard the ISS.		

Task Description:	2) A study on CH* chemiluminescence and soot pyrometry. This includes understanding the relationship between CH* emission and ion current. 3) Quantifying the risk of CO emission from quenched jet flames and identifying a flame's "electrical boundary." The boundary is defined as the location of a first detectable ion current. 4) Experiments designed to measure the ion current and the thermal emission of a flame as it is progressively quenched at a surface. Progress in these areas may improve kinetic mechanisms for modeling and help reduce the risks of CO poisoning from quenched flames. Task Description: The overall goal of this project is to understand the behavior of chemi-ionization in flames and to demonstrate the use of this understanding to manipulate combustion in beneficial ways (including enhancing flame stability and reducing flame soot). The project uses detailed measurements of electrical properties of flames in several ways. First, we can create a neutral wind (ion driven convection) locally at the flame front; this is one of the rare and important action-at-a-distance effects on flames that can be used as a localized actuator. The effect of the wind has been demonstrated with a candle flame in parabolic flight based microgravity and in a spherical cage drop tower, but there has been no systematic study of the ion wind effect in the absence of the confounding effects of the approximately equal influence of buoyancy. Second, we can use electric fields to affect soot formation and transport – ions and soot are closely linked because there are shared chemical pathways and because soot particles are charged by the flame ions. It is possible, therefore, to modify soot formation and to use charges attached to soot to affect soot transport. By adjusting the soot trajectory through the reaction zone it is possible to affect the oxidation of soot to prevent its eventual release or to modify the residence time of soot in different zones of the flame. It is also possible to use the neutral ion wind to change the mixing behavior of the flame and reduce soot in this way. Finally, we can use the electrical properties to characterize the flame – there is a quantifiable relationship between the carbon content of a fuel and the ion current it can produce at saturation. This relationship is already exploited in a typical flame ionization detector (FID) analyzer for hydrocarbon emission where the carbon is burned in a hydrogen flame to create an ion current in proportion to the carbon influx. In direct flame interrogation, however, there are several confounding variables that can affect the ion current, including flame temperature (which generally affects the O atom concentration) and the reaction zone volume. Flame temperature can in turn be affected by levels of premixing, diffusion rates, fuel composition, and radiative loss. It is possible, therefore, to determine flame character from the ionization curve response. Task Significance: Understanding electric field interactions with flames is central to combustion control, particularly in situations near limit operation of the flame where small effects are amplified dramatically. Electric fields have been shown to modify burning rates as well as to enhance and reduce flame propagation even to the point of flame extinguishment. Our experiments provide a unique ability to make simultaneous measurements of both physical and electrical properties of flames.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The control of combustion has the potential to improve efficiency and reduce emissions from burning fuels. Since high power density often requires combustion, these improvements will be important no matter what the fuel source. Electric fields acting on flames have the potential to aid in combustion control. In addition, electrical properties of flames can identify poor performing boiler flames that release poisonous carbon monoxide. Our studies show that a flame's electric signature can capture incipient quenching before dangerous emissions result. Understanding the links between electrical character and flame behavior may allow improved sensing of poor performing combustion systems.
	High Voltage Power Supply (HVPS) Study A portion of our research this year was directed toward characterizing a pair of High Voltage Power Supplies (HVPS) manufactured by UltraVolt. The combined supply is the proposed high voltage source for a future space flight experiment aboard the ISS. Although selected for its small size and specifications, it is necessary to validate the supply performance under the proposed spaceflight conditions. NASA provided the power supply circuit diagram. In order to understand how the HVPS controls a process, a simple, basic approach on its output response is used and compared to a high-performance TREK supply used in experiments at UC Irvine. The test consisted of applying a voltage step from -10kV to +10kV and measuring the time required for the output voltage to change between 10% and 90% of the requested voltage. The measured rise time of the UltraVolt supply (~85ms) is considerably longer than that TREK supply (~0.9 ms). It should be noted that both supplies have a short time delay of about 4ms from the time of the request to when the voltage begins to rise. This is a negligible effect and likely an error introduced by the data acquisition hardware. The results indicate that further evaluation is needed before we can accept the performance of the current power supplies. A follow-on study will investigate the system response for additional step and ramp functions, and an evaluation of noise. The effect of these non-ideal behaviors on the flame response will be evaluated. UCI Coflow Burner All ground-based experiments were performed using a coflowing burner that is dimensionally similar to the proposed ISS burner. The burner consists of two stainless steel plenums threaded together. Fuel, introduced in the bottom plenum, is passed through a stainless steel tube (2.13 mm ID) and exits the top surface fully developed. Air entering the top chamber passes through flow straightening beads and a honeycomb mesh. The air exits the top surface with a top hat

Task Progress:

flow profile at the exit. A variable high-tension power supply connected between the burner and downstream mesh electrode produces the applied electric field. The electric field distribution is deformed by the flame's space charge. Regardless, defining an applied field strength, E = V/d, is a useful approximation when describing experimental results. Ion current is computed using Ohm's law and the measured potential drop across a shunt resistor placed between the electrically isolated burner and the building ground. An acrylic chamber shields the flame from room currents.

Flame chemiluminescence and ion current

Results from a previous annual report discuss the strong relationship between the measured CH* chemiluminescence and the saturation ion current in non-sooting flames. The data suggested a bi-directional coupling between the luminous emission and the measured ion current. That is, the resultant ion wind modifies the flame structure, which influences ion production. This change in ion concentration intensifies the strength of the electrical body force and the measured ion current. At the saturation condition, the measured ion current can be related to the rate of ion production. However, because a flame must be altered by an electric field in order to measure the saturation current, a non-intrusive alternative may be possible using CH* chemiluminescence.

During the current reporting period we have expanded the conditions of the prior study to include sooting flames. In flames that produce incandescent soot, the emission spectrum from carbon, which is broad, interferes with CH* emission (~431 nm). In some cases, the soot spectrum collected through a narrowband filter centered at 430 nm contributed over 80% of the total emission measured. To filter out this noise, we investigated a method designed to remove the contribution of soot near the CH* emission line. The method involves measuring the emission through a pair of narrowband filters centered above and below this region. The scaled average of this measurement is proportional to the soot emission collected by a 430 nm filter to within 1%. The method was possible because of the near linear region in the broadband emission curve at 430 nm. After removing this contribution to the measurement, the remaining emission collected at 430 nm corresponds to CH*.

In lightly sooting methane flames, the method successfully removed the soot contribution from the flame image, and the resulting images reveal a previously obscured CH* distribution. In non-premixed ethylene flames, the method successfully removed the brightest portions of soot from the flame, leaving behind a luminous ring around the soot border. We originally suspected that this ring was an artifact of unsteadiness in the flame, but so far our careful repeated measurements suggest that this is not the case and that the signal does reflect CH*.

Because images are taken in sequence, this method is limited to steady flames. Follow on studies will explore tracking the CH* boundary in an unsteady sooting flame. If used with a calibrated PMT or a digital camera, this method may help determine the ratio of CH* radicals that eventually produce H_3O^+ ions, as well as identifying factors that influence the ratio. A calibration of the CH* emission was accomplished by Walsh et al., 2005, in determining the absolute CH* concentration (as compared to that of unexcited CH). We intend to use a similar approach.

Quenching studies

Professor Felix Weinberg at Imperial College is currently studying the effects of quenching on flame structure in order to predict the hazardous conditions created by poorly tuned industrial boilers. The ultimate practical aim of this series of experiments is to reduce the incidence of CO poisoning caused by the quenching of natural gas flames in malfunctioning burners. All the work involves progressively increasing proportions of air added to a constant fuel flow, so as to maintain a constant power input. The quenching measurements focus on compositions ranging from pure diffusion flames to progressively increasing air admixtures. The reason is that such burners (e.g., boilers) are designed to operate under near-stoichiometric conditions, and when they malfunction due, say, to an obstruction in the air supply, they tend to burn rich. Finally, we investigate how the flow of charge can be employed to signal the onset of hazardous conditions, before any CO release can occur.

Quenching

Concern about CO emission initially arose after observing appreciable concentrations of CO within the luminous boundaries of our flames. These results prompted the question of how much carbon monoxide would escape into the atmosphere if quenching disrupted the flame. We studied CO release for a variety of quenching surfaces, using an exhaust system which aspirated the emerging product gases, along with much ambient air.

The quenching bodies were clamped in a vernier positioner and the height of the quenching surface above the burner was recorded for each CO measurement. Of the numerous results, some show specific aspects of quenching behaviour. Although the Al grid, for example, also shows a small subsidiary peak that coincides with a maximum within the flame, it is clear that, close to the burner, quenching by some metals can generate CO from the reactants in excess of the amount present in the unperturbed flame. This is not surprising since Ni and Al have been used as constituents of catalysts for the production of syngas from natural gas. It implies that the use of new alloys in boiler tubing, and also metal impurities, may need to be scrutinized from that viewpoint.

We found no CO present from the products released beyond the region of luminosity which extends to about 16mm above the burner for diffusion flames and some 5 mm less for near stoichiometric premixed flames. This is worth bearing in mind, since all the parameters that follow are here plotted against H, the height above the burner.

The measurements imply that as much as 6% of the CH_4 feed may be converted to CO. This is equivalent to 3 – 5 L/min for the 30kW recommended for a 1 bedroom flat. During 1999–2004, CO poisoning caused 16,447 deaths in the United States, the principal cause being incomplete combustion in malfunctioning heaters/boilers.

Electrical detection of proximity of quenching surface

The facility of monitoring flame – solid surface interactions electrically may have a variety of applications, not necessarily linked to quenching. In the present context, however, our chief objective was its use to detect the proximity to the flame of a quenching surface, well before any hazard of CO escape could arise. All the aforegoing measurements indicate that such a warning device should be based on a negatively charged burner to utilize the farther electron penetration. Comparison of the results already shows that as a "quench detector" the current with the burner negative, signals the approach to a quenching surface at a height which is more than 30 mm in excess of that at which any quenching is ever recorded.

In practice, of course, quenching may occur by surfaces not directly above the flame. To explore the loci of electrical

quenching detection, a rod with a rounded end was used as the positive electrode. It was clamped in the x-y-z positioner and manipulated to approach the flame orthogonally at locations all around the flame periphery. The contours of first detection of a current for both a pure fuel diffusion flame and a near-stoichiometric premixed flame at two potentials (37 and 56 V) were determined. Interestingly, the contours are remarkably similar to each other, in spite of the circumstance that the flow rate in the latter is ten times greater. They lie massively outside the regions in which CO is seen or detected in the products of quenching.

We conclude that detecting the proximity of quenching surfaces from a flame, to alert the user before any carbon monoxide is released, is feasible and requires only the use of small batteries. The detection is particularly effective in the hot post-flame region of the flow. It could readily be developed into a system which would give warning of hazardous conditions arising, say, from an unexpected elongation or change in flame shape due to, for example, an obstruction in the feed.

Near quenching

In addition to our efforts at Imperial College, parallel studies are underway at UC Irvine, studying the electrical properties of partially quenched flames. Our work at Imperial College has illuminated the potential dangers of improperly tuned boilers and the high concentration of carbon monoxide and hydrogen potentially emitted when quenched. This portion of our research is aimed at understanding the electrical behavior of a flame as a plate is incrementally lowered toward it by measuring the corresponding ion current. The study is a proof of concept to explore the potential of detecting flame impingement electrically without visual inspection. Previous work has shown that ion current sharply increases and finally saturates when lowering the plate toward the jet flame exit. The ion current decreases as the flame temperature and ion production fall.

We continued the work to study the phenomena occurring on the quenching plate as the flame approached it. Different quenching surface materials have various emissivity and thermal properties that lead to different equilibrium temperatures. Preliminary research involves identifying the thermal emission from the plate at different heights when lowered towards the flame (or when the flame sheet is interrupted). In order to understand the thermal phenomenon, a FLIR camera is used. The FLIR camera uses a thermography technique for taking thermal images. The test ran with a stainless-steel plate that had been painted with a layer of non-gloss heat resistant paint to prevent reflections. The FLIR camera was located right above from the top view of the plate while a flame is quenched underneath the plate. The image shows the plate’s temperature varies between 42 and 208 C, and the average is 112 C, when the emissivity is set as 0.98 and there is no viewing angle difference from the observed position. The next phase of this work will involve measuring electrical properties and their relationship to carbon monoxide release near quenching.

Bibliography Type:	Description: (Last Updated: 11/26/2025)
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